Kai Sheng Tai

kst@cs.stanford.edu
https://kaishengtai.github.io
https://github.com/kaishengtai

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About

I am an AI researcher working on efficient neural network architectures and training methods. Most recently, I worked at Meta on developing on-device LLMs and on pretraining text encoder foundation models for recommendation and retrieval systems. My research has explored ways to leverage structure and invariances in data to build more efficient and effective machine learning models: for instance, by exploiting the structure of language with Tree-LSTMs, the various symmetries in images with Equivariant Transformers, and the underlying physical constraints of the Earth for ML-driven earthquake detection methods.

Education

2021 Ph.D. in Computer Science, Stanford University
Thesis: *Statistical Machine Learning Under Resource Constraints*
Advisors: Peter Bailis and Gregory Valiant

2015 M.S. in Computer Science, Stanford University

2013 A.B. in Physics, *magna cum laude*, Princeton University
Thesis: *Detecting Gravitational Waves from Highly Eccentric Compact Binaries*
Advisors: Frans Pretorius and Sean McWilliams

Professional Experience

2021–2025 Research Scientist, Meta
- *Meta Reality Labs:* Post-training for on-device LLMs, LLM quantization and pruning
- *Modern Recommendation Systems:* Technical lead for internal text encoder pre-training
- *Facebook AI Applied Research:* Algorithm design for sparse model training

2016–2021 Ph.D. Student, Stanford University

2015–2016 Senior Data Scientist, MetaMind (acquired by Salesforce in April 2016)
- Developed 3D convolutional networks for medical MRI classification

2014–2015 Research Assistant, Natural Language Processing Group, Stanford University

Selected Publications

Hanqing Zhu, Zhenyu Zhang, Hanxian Huang, DiJia Su, Zechun Liu, Jiawei Zhao, Igor Fedorov, Hamed Pirsiavash, Zhizhou Sha, Jinwon Lee, David Z Pan, Zhangyang Wang, Yuandong Tian, and **Kai Sheng Tai**. The Path Not Taken: RLVR Provably Learns Off the Principals. arXiv 2025.

Patrick Huber, Ernie Chang, Wei Wen, Igor Fedorov, Tarek Elgamal, Hanxian Huang, Naveen Suda, Chinnadhurai Sankar, Vish Vogeti, Yanghan Wang, Alex Gladkov, **Kai Sheng Tai**, Abdelrahman Elogeel, Tarek Hefny, Vikas Chandra, Ahmed Aly, Anuj Kumar, Raghuraman Krishnamoorthi, and Adithya Sagar. MobileLLM-Pro Technical Report. arXiv 2025.

A. Tuan Nguyen, **Kai Sheng Tai**, Sirius Chen, Satya Narayan Shukla, Hanchao Yu, Philip Torr, Taipeng Tian, and Ser-Nam Lim. uCAP: An Unsupervised Prompting Method for Vision-Language Models. ECCV 2024 (oral presentation).

Kai Sheng Tai, Taipeng Tian, and Ser-Nam Lim. Spartan: Differentiable Sparsity via Regularized Transportation. NeurIPS 2022.

Weiqiang Zhu*, **Kai Sheng Tai***, S. Mostafa Mousavi, Peter Bailis, and Gregory C. Beroza. An End-to-End Earthquake Monitoring Method for Joint Earthquake Detection and Association using Deep Learning. *Journal of Geophysical Research: Solid Earth*, 2022. (*equal contribution)

Kai Sheng Tai, Peter Bailis, and Gregory Valiant. Sinkhorn Label Allocation: Semi-Supervised Classification via Annealed Self-Training. ICML 2021.

Kai Sheng Tai, Peter Bailis, and Gregory Valiant. Equivariant Transformer Networks. ICML 2019.

Vatsal Sharan*, **Kai Sheng Tai***, Peter Bailis, and Gregory Valiant. Compressed Factorization: Fast and Accurate Low-Rank Factorization of Compressively-Sensed Data. ICML 2019. (*equal contribution)

Edward Gan, Jialin Ding, **Kai Sheng Tai**, Vatsal Sharan, and Peter Bailis. Moment-Based Quantile Sketches for Efficient High Cardinality Aggregation Queries. VLDB 2018.

Kai Sheng Tai, Vatsal Sharan, Peter Bailis, and Gregory Valiant. Sketching Linear Classifiers over Data Streams. SIGMOD 2018.

Kai Sheng Tai, Richard Socher, and Christopher D. Manning. Improved Semantic Representations from Tree-Structured Long Short-Term Memory Networks. ACL 2015.